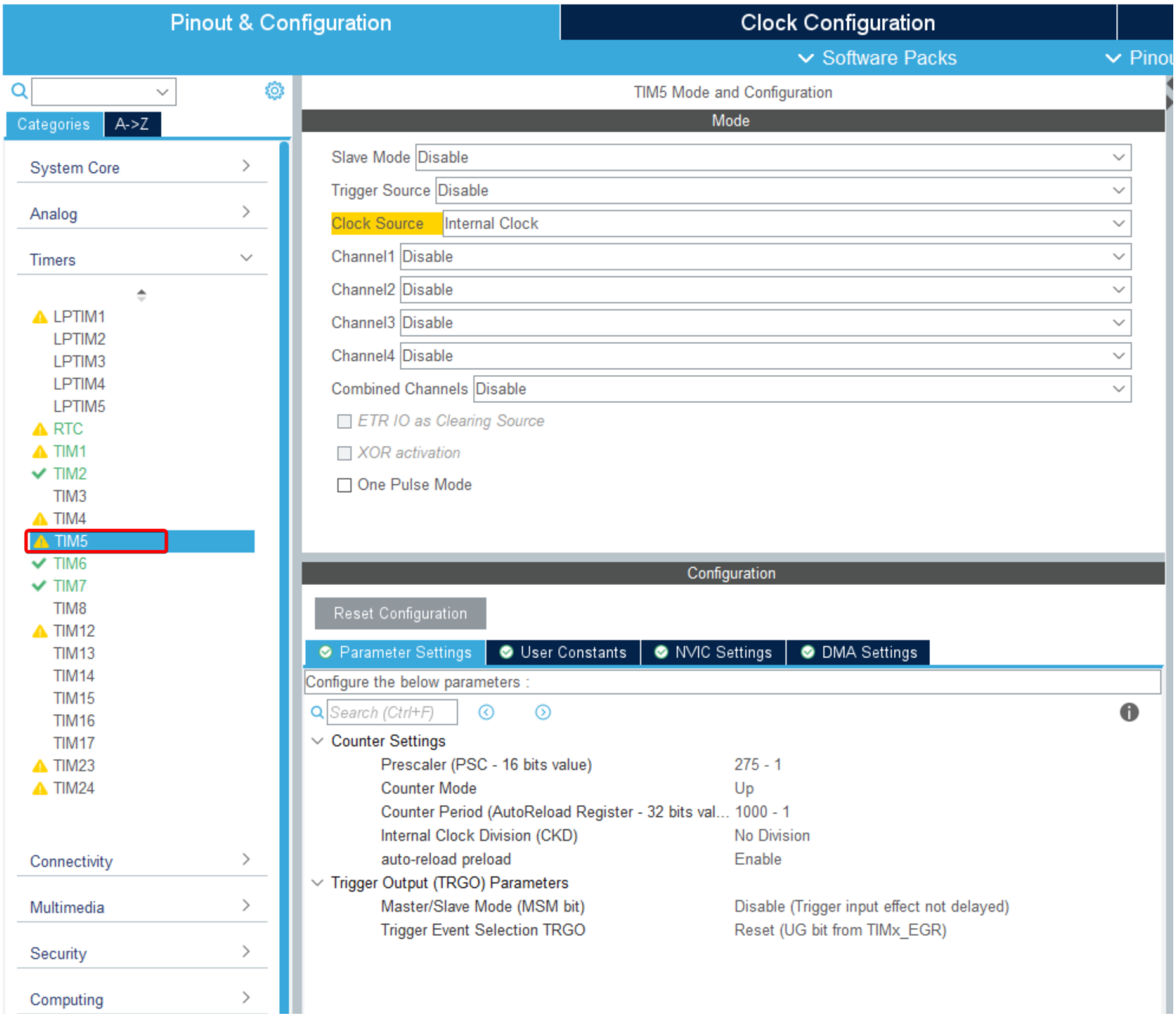
1. 文档说明

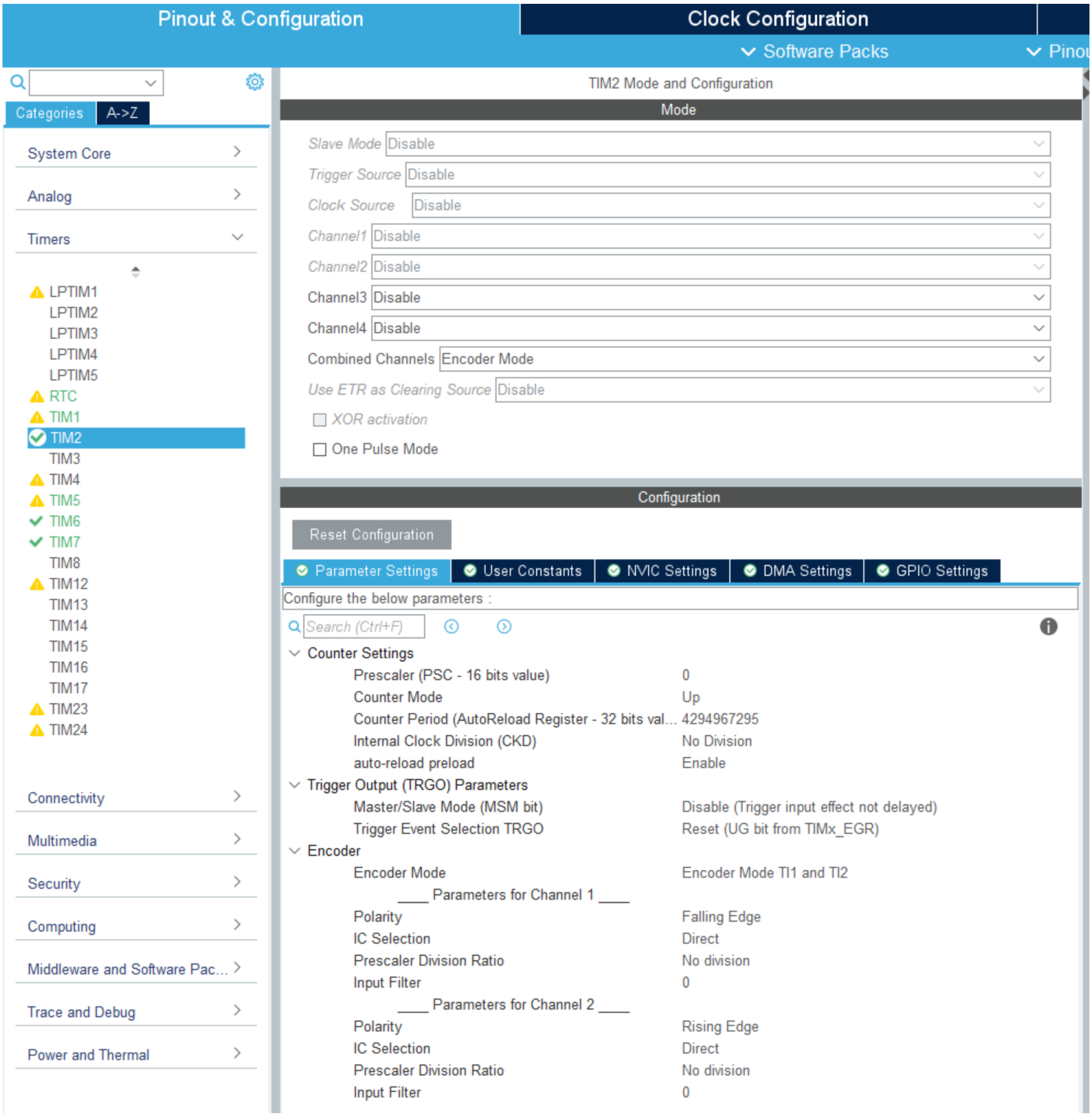
- 本demo针对业务变更情况，此时需要更改片上外设功能配置，并需要编写设备驱动、接口程序
- 本demo拟实现如下功能：
  - 将`ethercat`周期性`ECAT_CheckTimer()`功能的时基由`TIM2`变更为`TIM5`
  - 将`TIM2`配置为编码器模式，对外部编码器脉冲进行计数
  - 基础分支：a13a11836aeb6d78a91e6443d3cc124d17173a15
  - demo分支：5768bb37671aadfa51188df45e0a8719115772ad

2. 修改方式

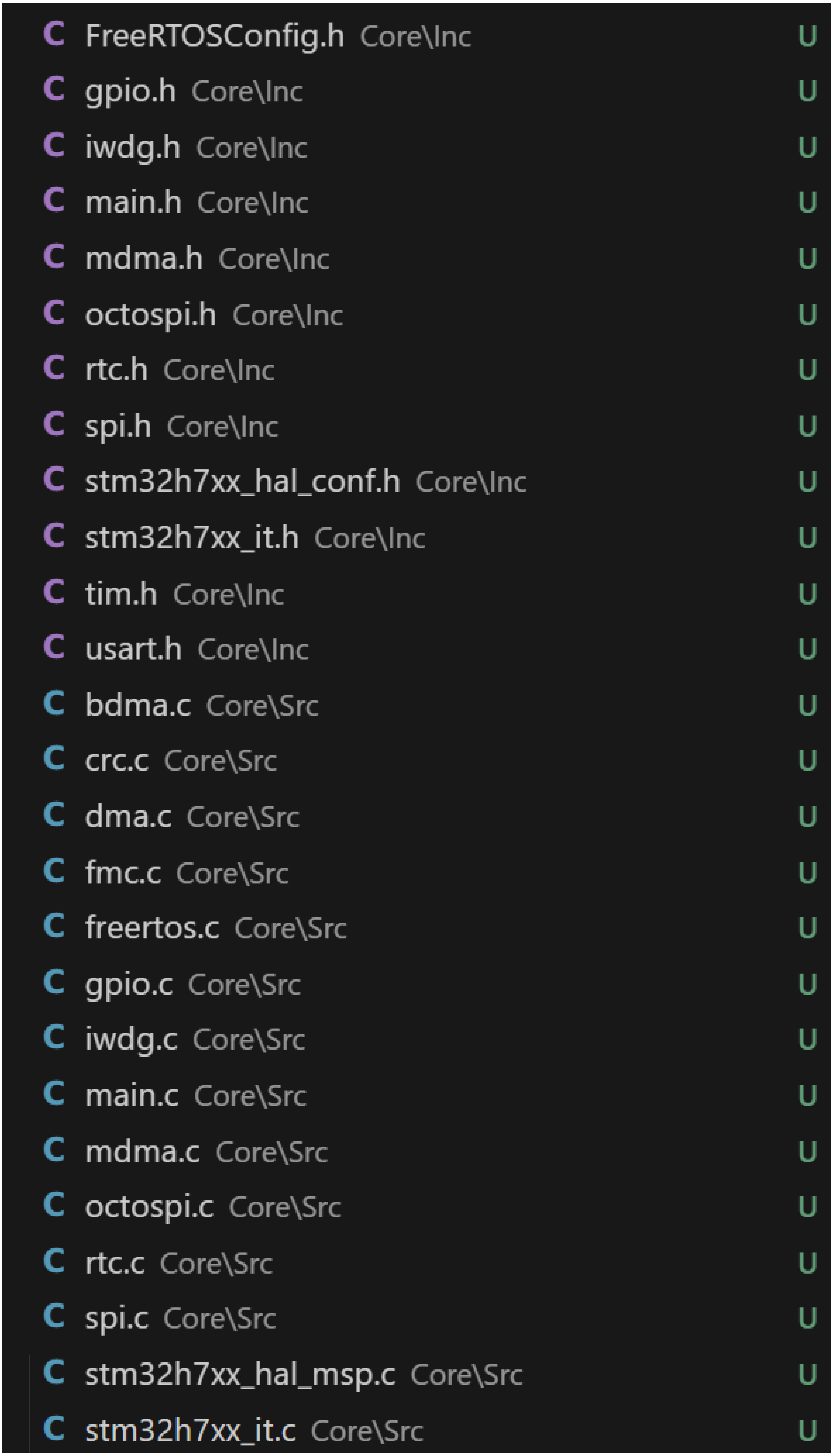
- 配置`TIM5`，参数与`TIM2`保持一致

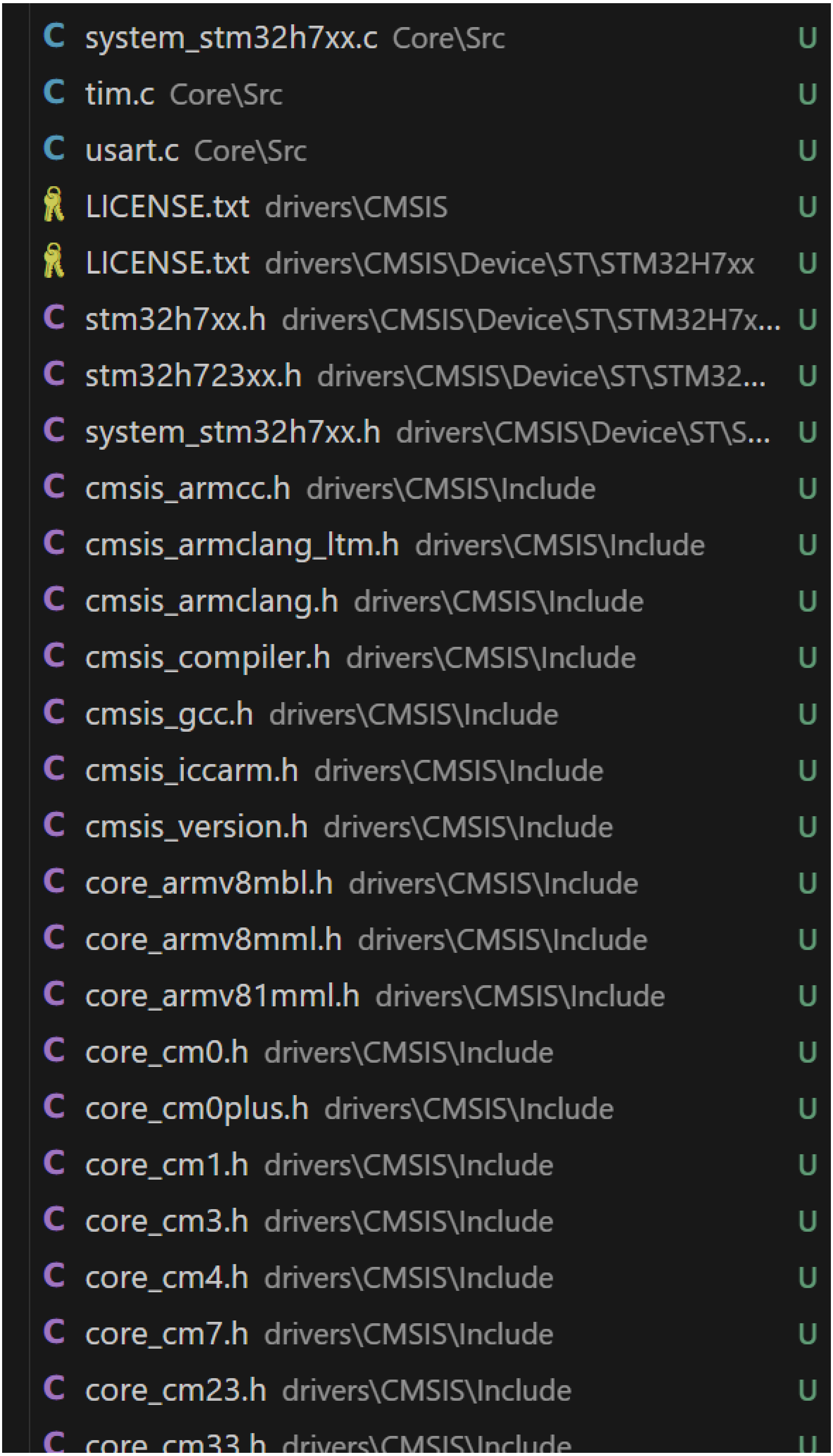


- 修改TIM2为编码器模式，并配置编码器输入引脚为PA0和PA1

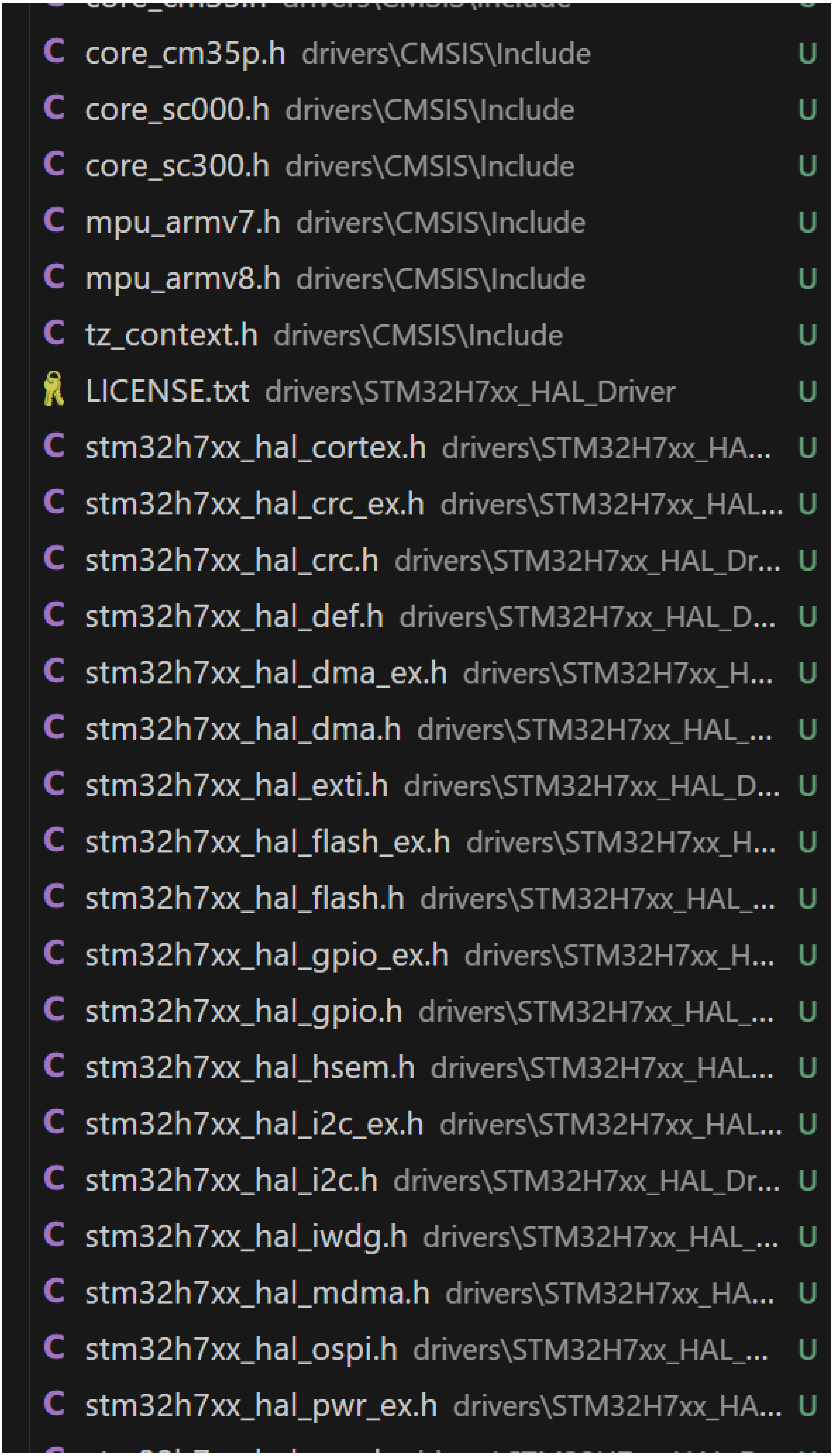


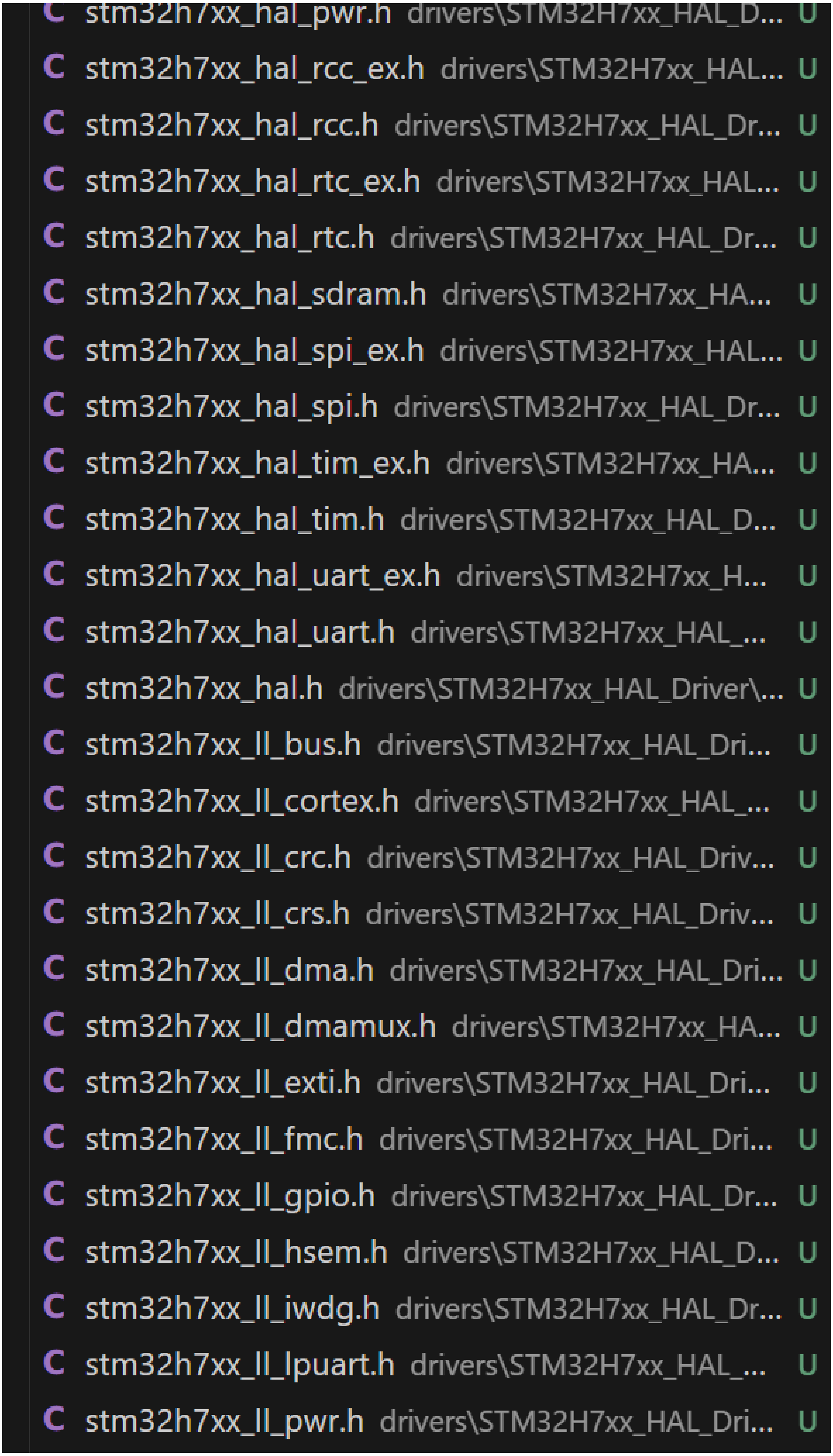
- STM32CubeMX重新生成代码，在使用GIT管理工程项目时，可看到新生成的工程相对于原工程有149个文件变更。在工程目录下，可发现新增/更改的文件或文件夹后面有**点状**或者**M/U/D**字符样式标识，即表示相关文件或目录下有新增/修改内容

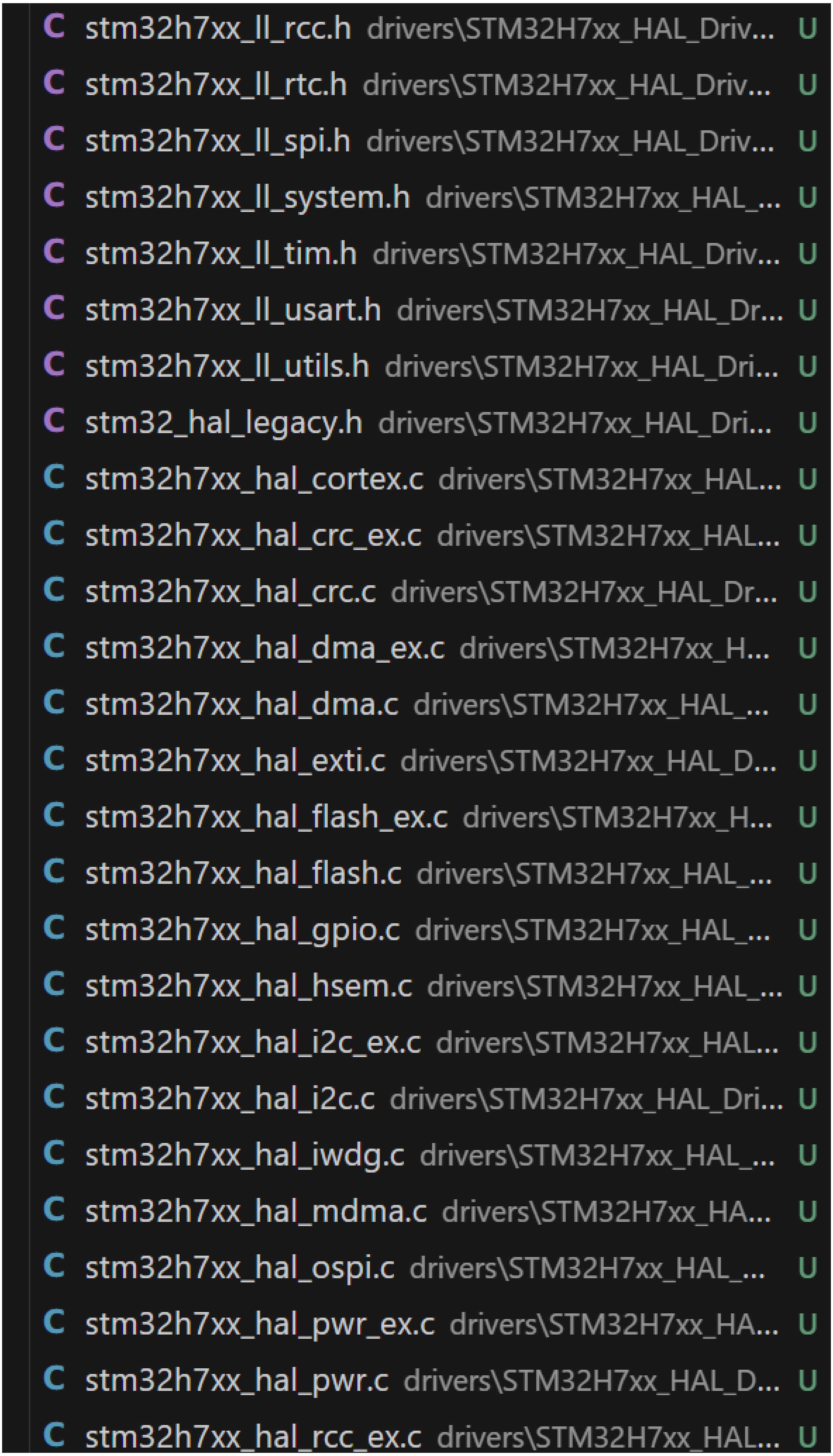

C FreeRTOSConfig.h Core\Inc U
C gpio.h Core\Inc U
C iwdg.h Core\Inc U
C main.h Core\Inc U
C mdma.h Core\Inc U
C octospi.h Core\Inc U
C rtc.h Core\Inc U
C spi.h Core\Inc U
C stm32h7xx_hal_conf.h Core\Inc U
C stm32h7xx_it.h Core\Inc U
C tim.h Core\Inc U
C usart.h Core\Inc U
C bdma.c Core\Src U
C crc.c Core\Src U
C dma.c Core\Src U
C fmc.c Core\Src U
C freertos.c Core\Src U
C gpio.c Core\Src U
C iwdg.c Core\Src U
C main.c Core\Src U
C mdma.c Core\Src U
C octospi.c Core\Src U
C rtc.c Core\Src U
C spi.c Core\Src U
C stm32h7xx_hal_msp.c Core\Src U
C stm32h7xx_it.c Core\Src U

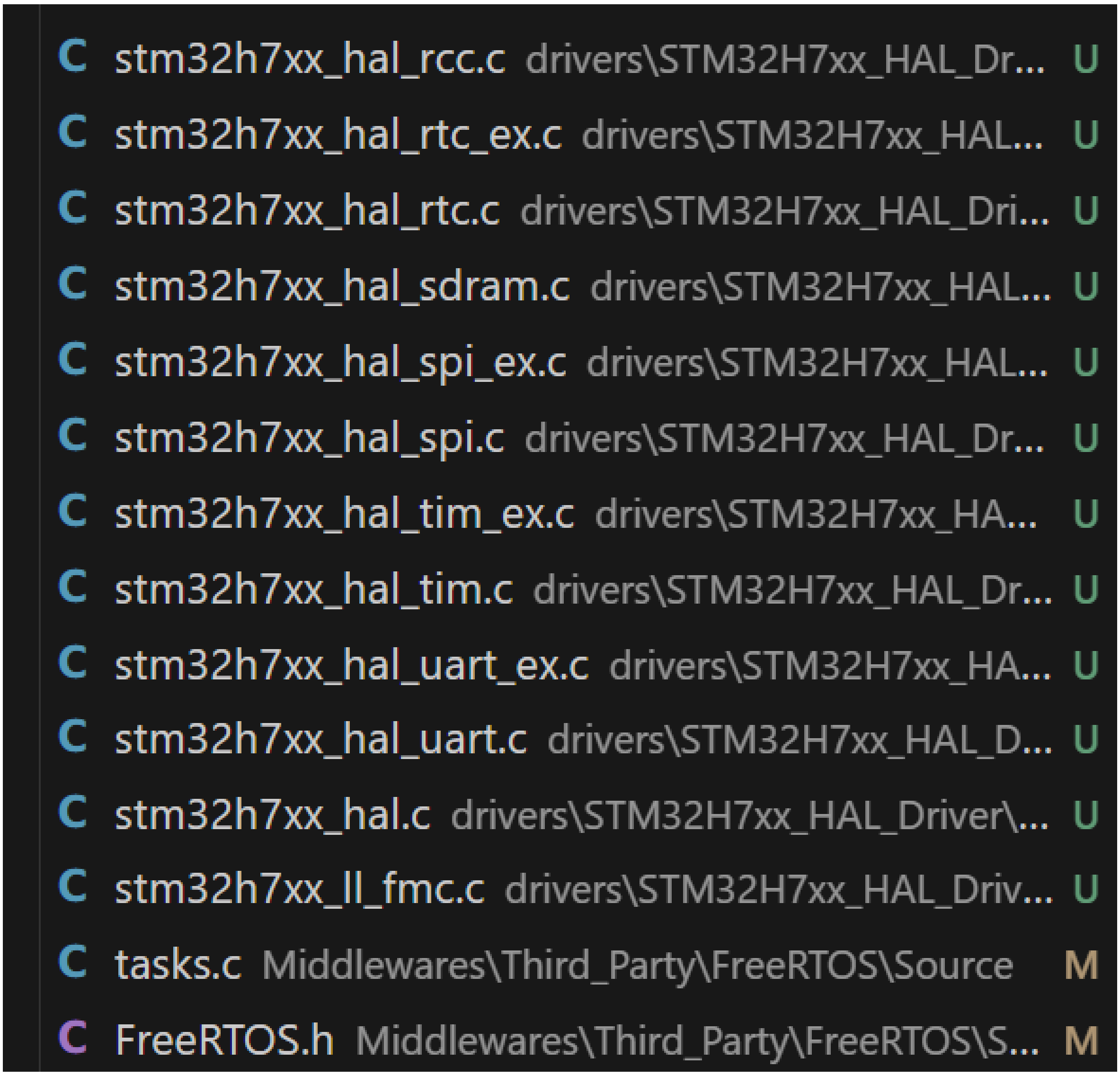


system_stm32h7xx.c Core\Src U
tim.c Core\Src U
usart.c Core\Src U
LICENSE.txt drivers\CMSIS U
LICENSE.txt drivers\CMSIS\Device\ST\STM32H7xx U
stm32h7xx.h drivers\CMSIS\Device\ST\STM32H7x... U
stm32h723xx.h drivers\CMSIS\Device\ST\STM32... U
system_stm32h7xx.h drivers\CMSIS\Device\ST\S... U
cmsis_armcc.h drivers\CMSIS\Include U
cmsis_armclang_ltm.h drivers\CMSIS\Include U
cmsis_armclang.h drivers\CMSIS\Include U
cmsis_compiler.h drivers\CMSIS\Include U
cmsis_gcc.h drivers\CMSIS\Include U
cmsis_iccarm.h drivers\CMSIS\Include U
cmsis_version.h drivers\CMSIS\Include U
core_armv8mbl.h drivers\CMSIS\Include U
core_armv8mml.h drivers\CMSIS\Include U
core_armv81mml.h drivers\CMSIS\Include U
core_cm0.h drivers\CMSIS\Include U
core_cm0plus.h drivers\CMSIS\Include U
core_cm1.h drivers\CMSIS\Include U
core_cm3.h drivers\CMSIS\Include U
core_cm4.h drivers\CMSIS\Include U
core_cm7.h drivers\CMSIS\Include U
core_cm23.h drivers\CMSIS\Include U

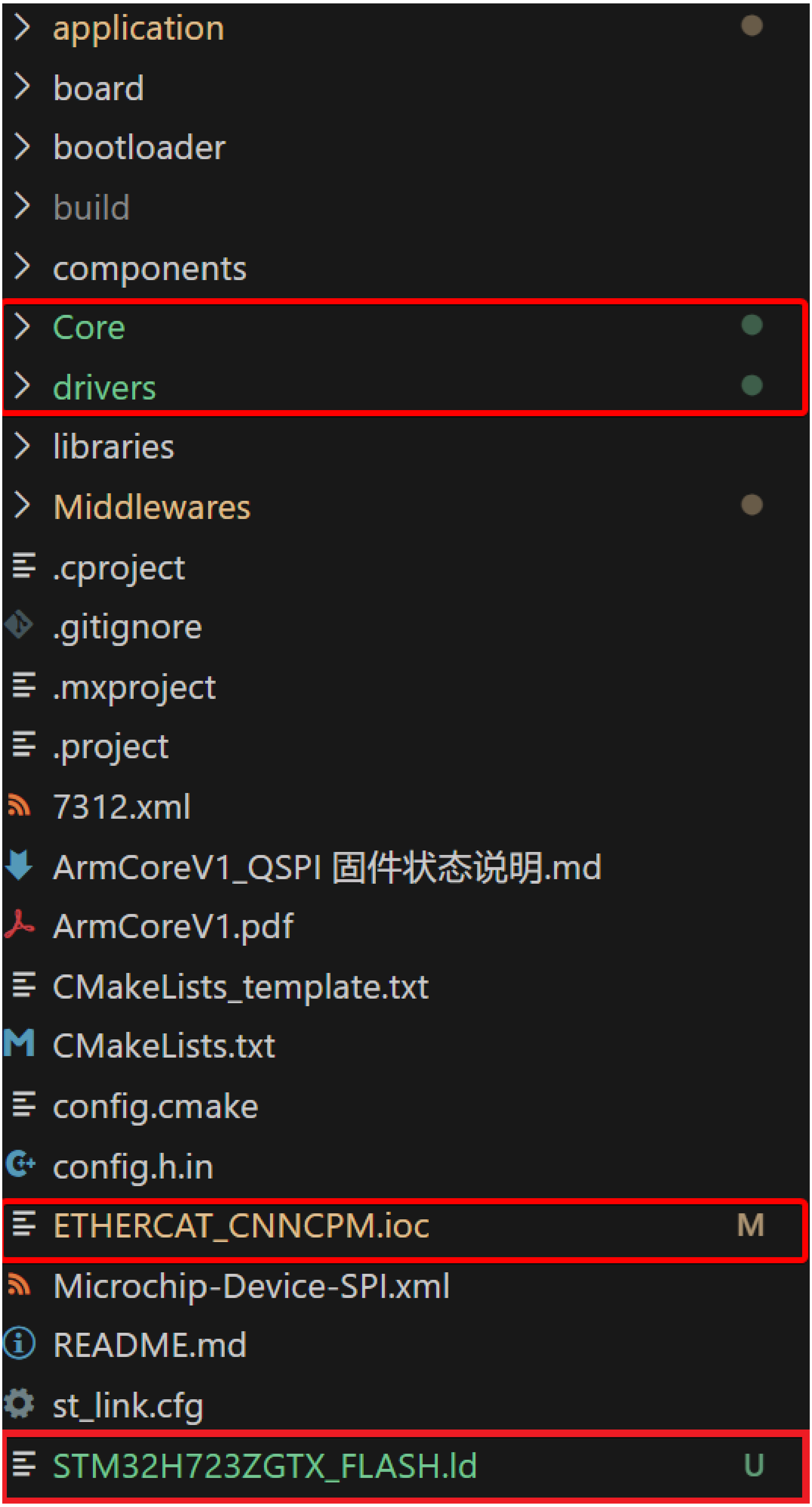

core_cm35p.h drivers\CMSIS\Include U
core_sc000.h drivers\CMSIS\Include U
core_sc300.h drivers\CMSIS\Include U
mpu_armv7.h drivers\CMSIS\Include U
mpu_armv8.h drivers\CMSIS\Include U
tz_context.h drivers\CMSIS\Include U
LICENSE.txt drivers\STM32H7xx_HAL_Driver U
stm32h7xx_hal_cortex.h drivers\STM32H7xx_HA... U
stm32h7xx_hal_crc_ex.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_crc.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_def.h drivers\STM32H7xx_HAL_D... U
stm32h7xx_hal_dma_ex.h drivers\STM32H7xx_H... U
stm32h7xx_hal_dma.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal_exti.h drivers\STM32H7xx_HAL_D... U
stm32h7xx_hal_flash_ex.h drivers\STM32H7xx_H... U
stm32h7xx_hal_flash.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal_gpio_ex.h drivers\STM32H7xx_H... U
stm32h7xx_hal_gpio.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal_hsem.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_i2c_ex.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_i2c.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_iwdg.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal_mdma.h drivers\STM32H7xx_HA... U
stm32h7xx_hal_ospi.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal_pwr_ex.h drivers\STM32H7xx_HA... U


stm32h7xx_hal_pwr.h drivers\STM32H7xx_HAL_D... U
stm32h7xx_hal_rcc_ex.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_rcc.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_rtc_ex.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_rtc.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_sdram.h drivers\STM32H7xx_HA... U
stm32h7xx_hal_spi_ex.h drivers\STM32H7xx_HAL... U
stm32h7xx_hal_spi.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_tim_ex.h drivers\STM32H7xx_HA... U
stm32h7xx_hal_tim.h drivers\STM32H7xx_HAL_D... U
stm32h7xx_hal_uart_ex.h drivers\STM32H7xx_H... U
stm32h7xx_hal_uart.h drivers\STM32H7xx_HAL_... U
stm32h7xx_hal.h drivers\STM32H7xx_HAL_Driver\... U
stm32h7xx_ll_bus.h drivers\STM32H7xx_HAL_Dri... U
stm32h7xx_ll_cortex.h drivers\STM32H7xx_HAL_... U
stm32h7xx_ll_crc.h drivers\STM32H7xx_HAL_Driv... U
stm32h7xx_ll_crs.h drivers\STM32H7xx_HAL_Driv... U
stm32h7xx_ll_dma.h drivers\STM32H7xx_HAL_Dri... U
stm32h7xx_ll_dmamux.h drivers\STM32H7xx_HA... U
stm32h7xx_ll_exti.h drivers\STM32H7xx_HAL_Dri... U
stm32h7xx_ll_fmc.h drivers\STM32H7xx_HAL_Dri... U
stm32h7xx_ll_gpio.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_ll_hsem.h drivers\STM32H7xx_HAL_D... U
stm32h7xx_ll_iwdg.h drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_ll_lpuart.h drivers\STM32H7xx_HAL_... U
stm32h7xx_ll_pwr.h drivers\STM32H7xx_HAL_Dri... U

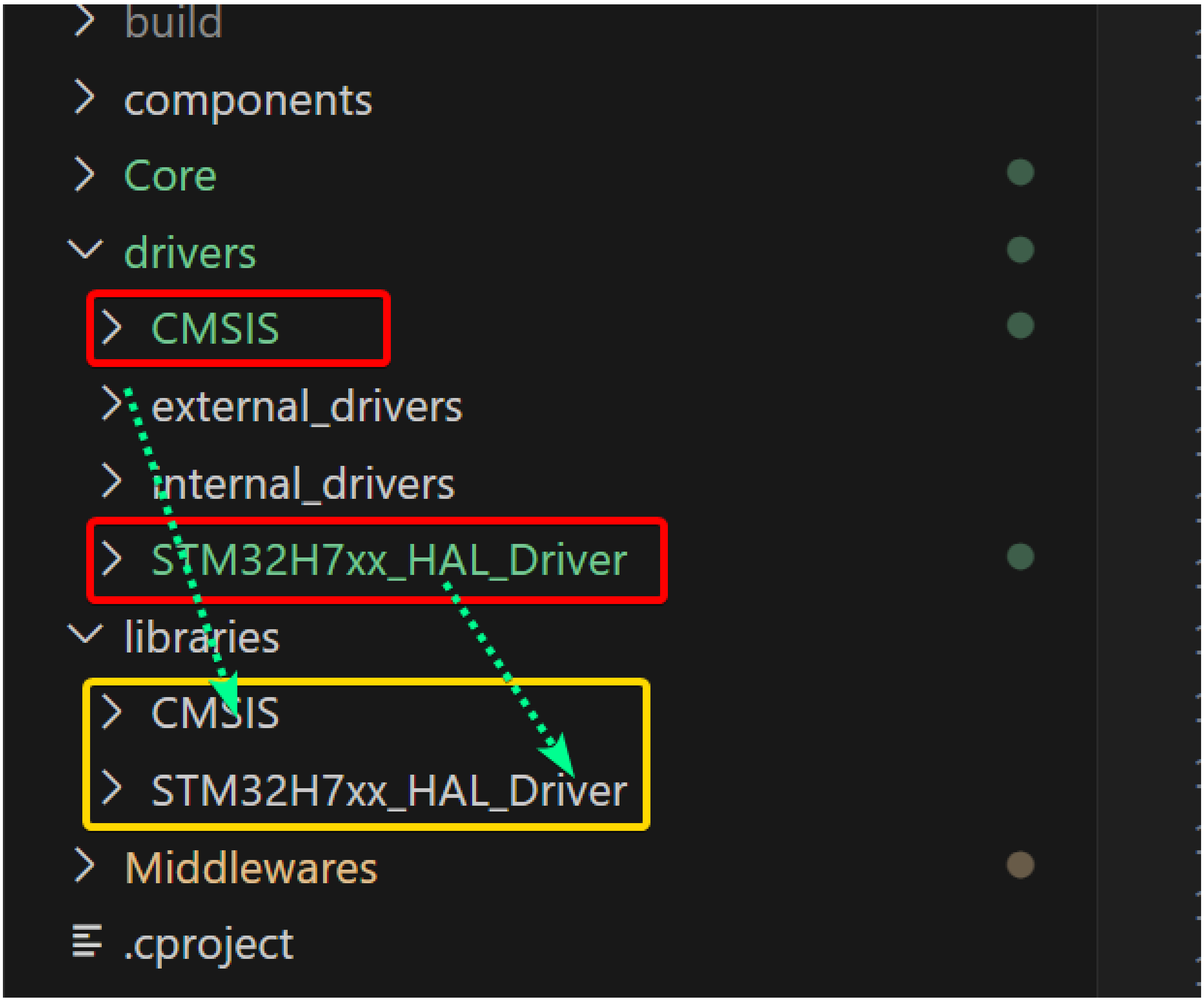

C stm32h7xx_ll_rcc.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_rtc.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_spi.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_system.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_ll_tim.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_usart.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_ll_utils.h drivers\STM32H7xx_HAL_Dri... U
C stm32_hal_legacy.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_hal_cortex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_crc_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_crc.c drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_dma_ex.c drivers\STM32H7xx_H... U
C stm32h7xx_hal_dma.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_exti.c drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_flash_ex.c drivers\STM32H7xx_H... U
C stm32h7xx_hal_flash.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_gpio.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_hsem.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_i2c_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_i2c.c drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_hal_iwdg.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_mdma.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_ospi.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_pwr_ex.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_pwr.c drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_rcc_ex.c drivers\STM32H7xx_HAL... U

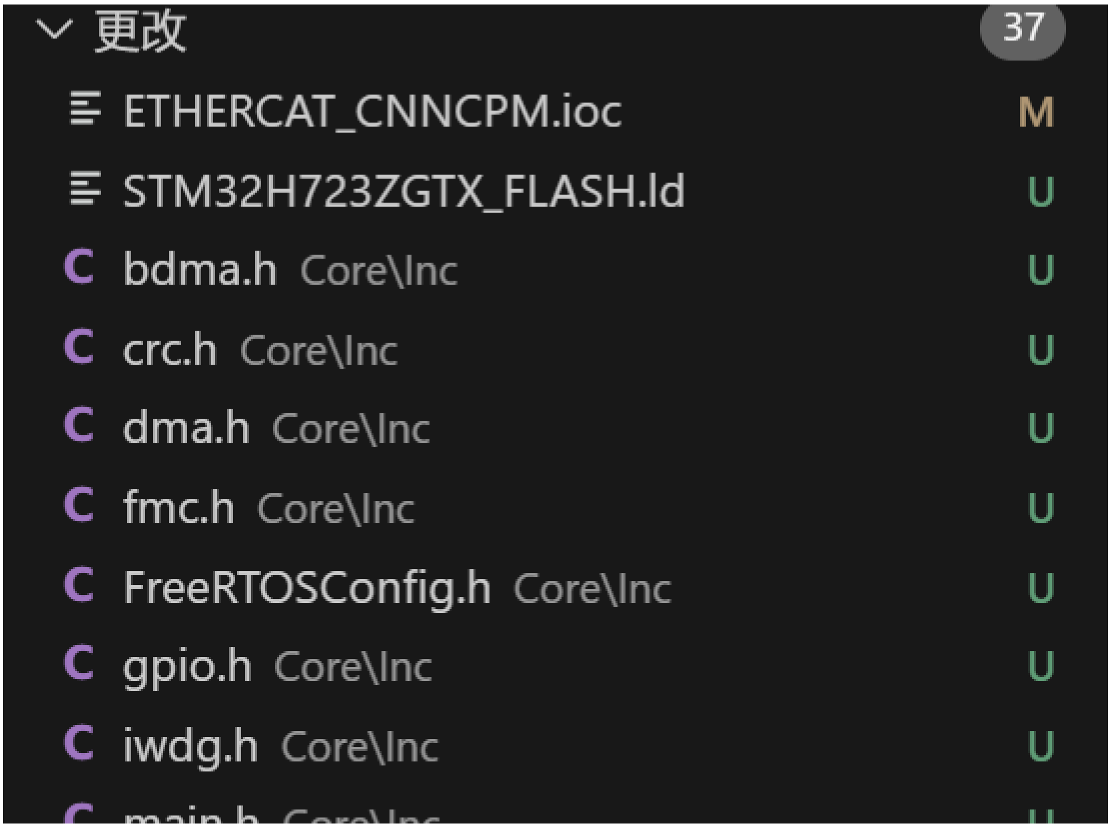


stm32h7xx_hal_rcc.c drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_rtc_ex.c drivers\STM32H7xx_HAL... U
stm32h7xx_hal_rtc.c drivers\STM32H7xx_HAL_Dri... U
stm32h7xx_hal_sdram.c drivers\STM32H7xx_HAL... U
stm32h7xx_hal_spi_ex.c drivers\STM32H7xx_HAL... U
stm32h7xx_hal_spi.c drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_tim_ex.c drivers\STM32H7xx_HA... U
stm32h7xx_hal_tim.c drivers\STM32H7xx_HAL_Dr... U
stm32h7xx_hal_uart_ex.c drivers\STM32H7xx_HA... U
stm32h7xx_hal_uart.c drivers\STM32H7xx_HAL_D... U
stm32h7xx_hal.c drivers\STM32H7xx_HAL_Driver\... U
stm32h7xx_ll_fmc.c drivers\STM32H7xx_HAL_Driv... U
tasks.c Middlewares\Third_Party\FreeRTOS\Source M
FreeRTOS.h Middlewares\Third_Party\FreeRTOS\S... M

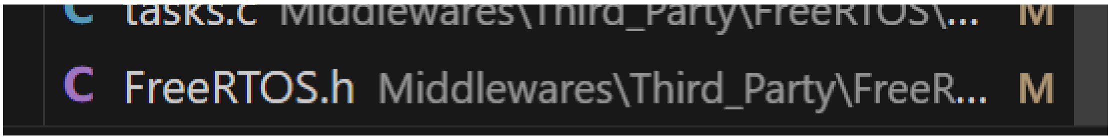

application
board
bootloader
build
components
Core
drivers
libraries
Middlewares
.cproject
.gitignore
.mxproject
.project
7312.xml
ArmCoreV1_QSPI 固件状态说明.md
ArmCoreV1.pdf
CMakeLists_template.txt
CMakeLists.txt
config.cmake
config.h.in
ETHERCAT_CNNCPM.ioc
M
Microchip-Device-SPI.xml
README.md
st_link.cfg
STM32H723ZGTX_FLASH.ld
U

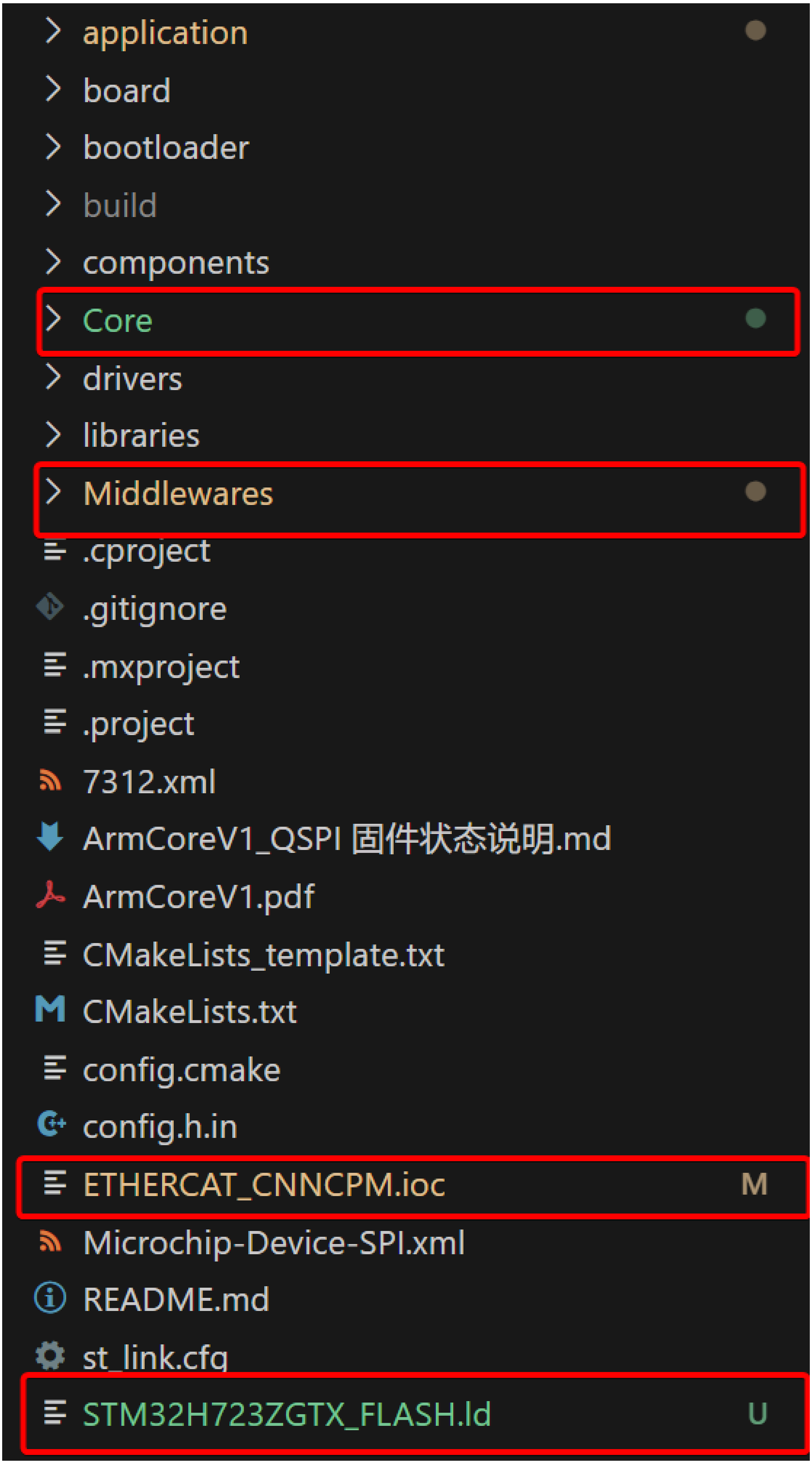
- 将工程`drivers`目录下`CMSIS`和`STM32H7xx_HAL_Driver`文件夹剪切到`libraries`目录下，直接合并替换`libraries`目录下原有的文件。经上述操作后，可看到差异文件减少为37个。同时，通过工程目录，新增/更改的文件分布在`Core`文件夹和`Middlewares`文件夹下，以及源文件`ETHERCAT_CNNCPM.ioc`、`STM32H723ZGTX_FLASH.ld`。

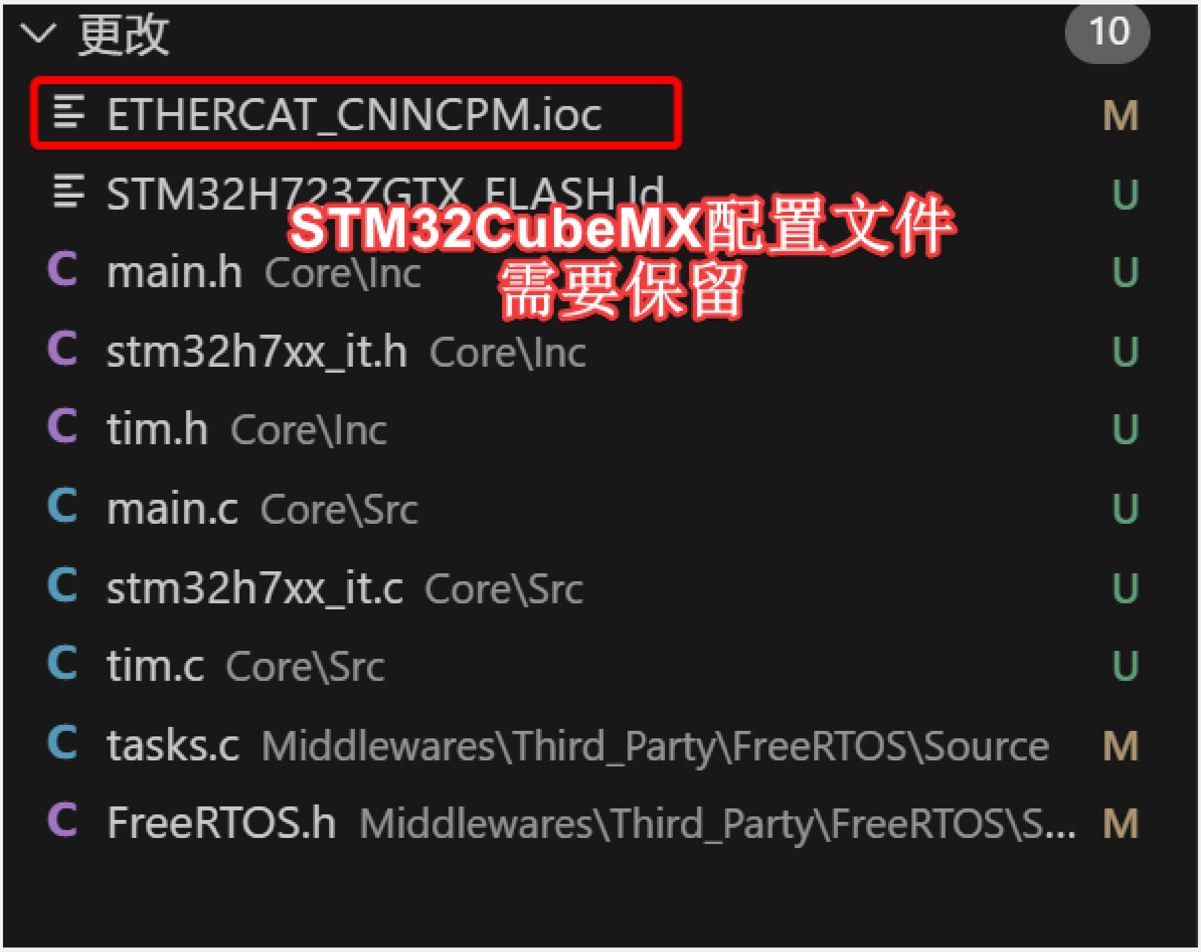





C mdma.h Core\Inc U
C octospi.h Core\Inc U
C rtc.h Core\Inc U
C spi.h Core\Inc U
C stm32h7xx_hal_conf.h Core\Inc U
C stm32h7xx_it.h Core\Inc U
C tim.h Core\Inc U
C usart.h Core\Inc U
C bdma.c Core\Src U
C crc.c Core\Src U
C dma.c Core\Src U
C fmc.c Core\Src U
C freertos.c Core\Src U
C gpio.c Core\Src U
C iwdg.c Core\Src U
C main.c Core\Src U
C mdma.c Core\Src U
C octospi.c Core\Src U
C rtc.c Core\Src U
C spi.c Core\Src U
C stm32h7xx_hal_msp.c Core\Src U
C stm32h7xx_it.c Core\Src U
C system_stm32h7xx.c Core\Src U
C tim.c Core\Src U
C usart.c Core\Src U

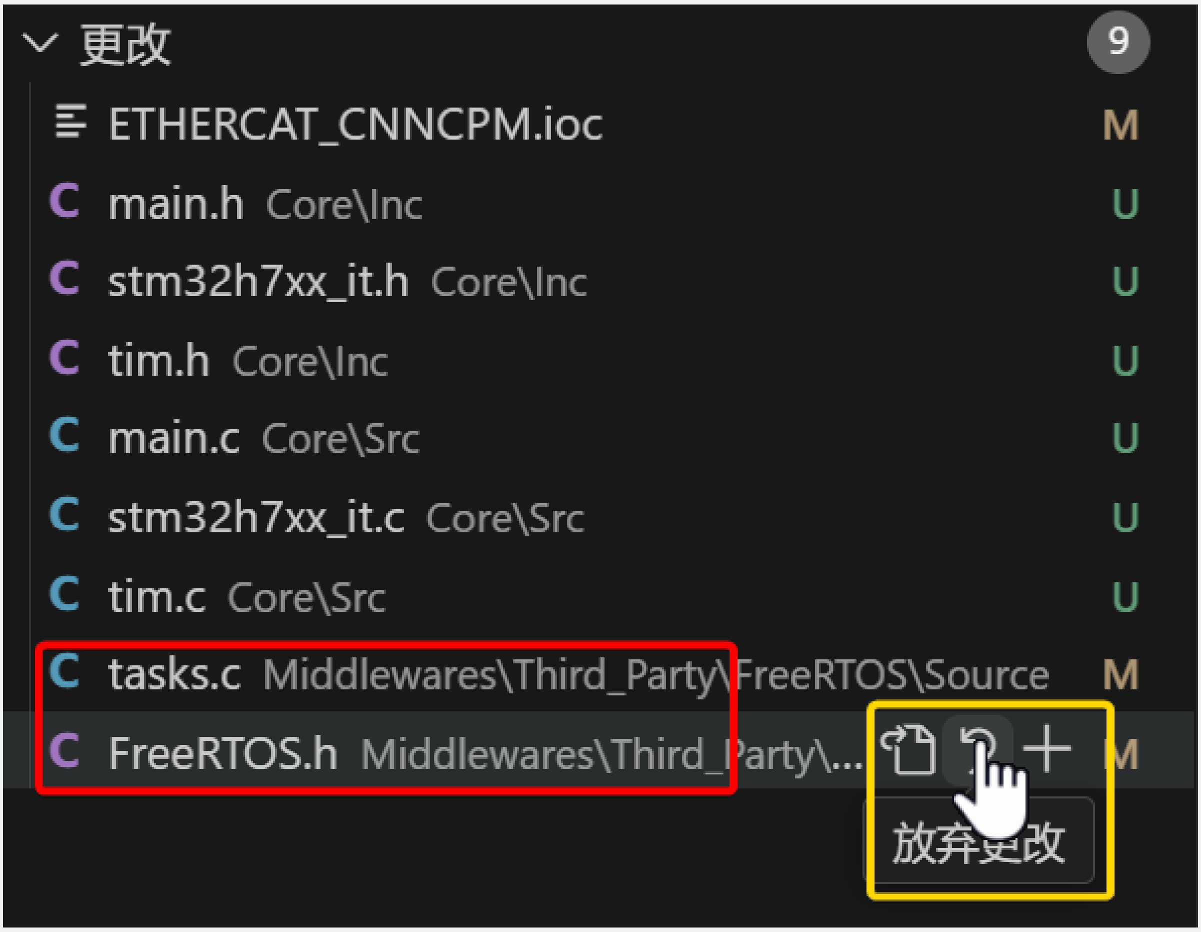

tasks.c Middlewares\Third_Party\FreeRTOS\... M
FreeRTOS.h Middlewares\Third_Party\FreeR... M

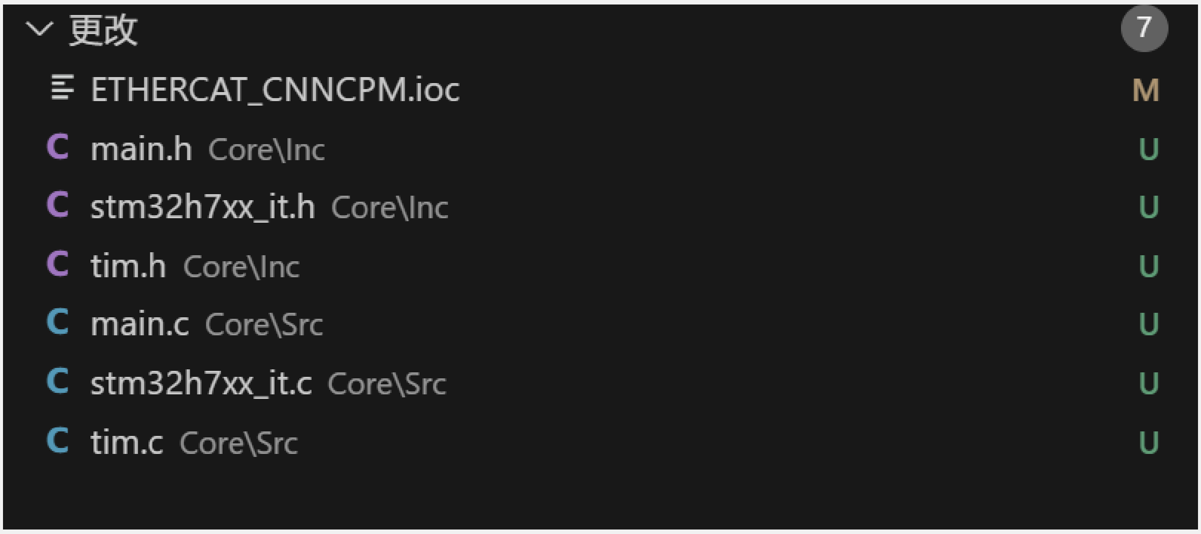

application
board
bootloader
build
components
Core
drivers
libraries
Middlewares
.cproject
.gitignore
.mxproject
.project
7312.xml
ArmCoreV1_QSPI 固件状态说明.md
ArmCoreV1.pdf
CMakeLists_template.txt
CMakeLists.txt
config.cmake
config.h.in
ETHERCAT_CNNCPM.ioc
M
Microchip-Device-SPI.xml
README.md
st_link.cfg
STM32H723ZGTX_FLASH.ld
U

> `ETHERCAT_CNNCPM.ioc`文件是`STM32CubeMX`的配置文件，记录了硬件的相关配置，**必须保留**，且每次更新硬件配置时均需要更新工程下该文件。
>
> 
>

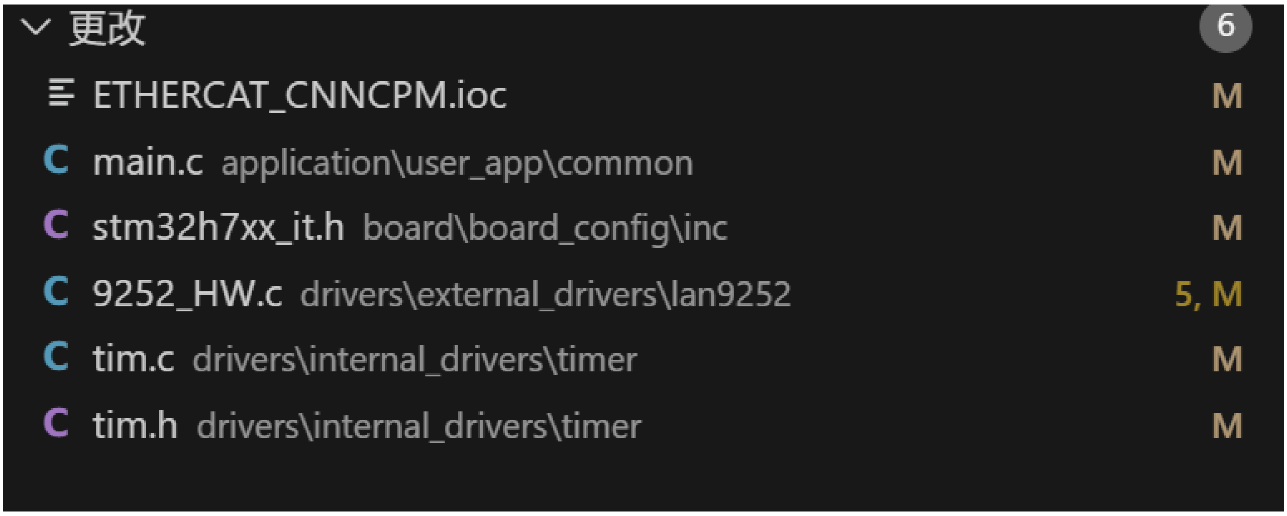
> `STM32H723ZGTX_FLASH.ld`文件是`STM32CubeMX`的链接脚本文件，需要将其与`board/linker_scripts`目录下的同名文件进行对比，并根据实际情况进行修改。因本demo不涉及链接脚本的修改，故直接删除该文件。

`Middlewares`是`STM32CubeMX`的`Middleware and Software Packs`分类菜单生成的中间件文件夹，当前仅放置了`FreeRTOS`相关文件。由于工程中移植了`cm_backtrace`组件用于硬件fault追踪。故，需要保留原文件（在相应文件上点击放弃更改，可直接删除更新的内容）。



> 将`Core`目录下，`Inc`和`Src`文件夹中的文件，与`drivers/internal_drivers`文件夹下同名文件进行对比，并修改`drivers/internal_drivers`文件夹下的对应文件内容。本demo只涉及timer的变更，故只需对比修改`tim.h`、`tim.c`、`main.h`、`main.c`、`stm32h7xx_it.h`和`stm32h7xx_it.c`这几个文件。将`Core`文件夹下未涉及文件删除后，如下：
>
> 
>

- 修改完成后的文件更改情况



- 若`CmakeLists.txt`文件有修改，则需要重新配置工程（重新生成makefile文件）

- 编译工程

```
问题 13   输出   调试控制台   终端   端口   MEMORY   XRTOS   GITLENS

[ 91%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_sdram.c.obj
[ 92%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi.c.obj
[ 93%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi_ex.c.obj
[ 94%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim_ex.c.obj
[ 96%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart.c.obj
[ 97%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart_ex.c.obj
[ 98%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_ll_fmc.c.obj
[100%] Linking C executable ETHERCAT_CNNCPM.elf
Memory region         Used Size  Region Size  %age Used
         ITCMRAM:           0 GB        64 KB      0.00%
         DTCMRAM:           0 GB       128 KB      0.00%
           FLASH:       163248 B         1 MB     15.57%
          RAM_D1:       129624 B       320 KB     39.56%
          RAM_D2:           1 KB        32 KB      3.13%
          RAM_D3:          176 B        16 KB      1.07%
         RAM_BKP:            6 B         4 KB      0.15%
Building F:/WORK/arm_core_develop/arm_core_restruction/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.hex
Building F:/WORK/arm_core_develop/arm_core_restruction/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.bin
[100%] Built target ETHERCAT_CNNCPM.elf
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build>
```

- 下载程序测试

```
问题 19   输出   调试控制台   终端   端口   MEMORY   XRTOS   GITLENS

Info : STLINK V2J40S7 (API v2) VID:PID 0483:3748
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build> make flash
Open On-Chip Debugger 0.12.0 (2023-10-02) [https://github.com/sysprogs/openocd]
Licensed under GNU GPL v2
libusb1 09e75e98b4d9ea7909e8837b7a3f00dda4589dc3
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use 'transport select <transport>'.
Info : The selected transport took over low-level target control. The results might differ compared to plain JTAG/SWD
Info : clock speed 1800 kHz
Info : STLINK V2J40S7 (API v2) VID:PID 0483:3748
Info : Target voltage: 3.207737
Info : [stm32h7x.cpu0] Cortex-M7 r1p2 processor detected
Info : [stm32h7x.cpu0] target has 8 breakpoints, 4 watchpoints
Info : starting gdb server for stm32h7x.cpu0 on 3333
Info : Listening on port 3333 for gdb connections
[stm32h7x.cpu0] halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x08006b48 msp: 0x24050000
** Programming Started **
Info : Device: STM32H72x/73x
Info : flash size probed value 1024k
Info : STM32H7 flash has a single bank
Info : Bank (0) size is 1024 kb, base address is 0x08000000
Info : Padding image section 1 at 0x08027db0 with 16 bytes (bank write end alignment)
Warn : Adding extra erase range, 0x08027dc0 .. 0x0803ffff
Warn : no flash bank found for address 0x38000000
Warn : no flash bank found for address 0x38800000
** Programming Finished **
** Resetting Target **
shutdown command invoked
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build>
```

- 至此，完成硬件配置的修改，下一步根据业务需求，编写相应的设备驱动接口。本demo编码器接口实现可参考**SHA-1: b5edeeab0f16bb55d3f87af913d04df3caab39ee**下`encoder_port.h`和`encoder_port.c`文件